🌍 PHYSICS CH.7 — GRAVITATION — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this chapter:

1. Mass vs Weight: Moon par Mass Same, Weight 1/6

- Mass = intrinsic property; kahin bhi same rehti hai (Earth, Moon, space sab jagah). *[Q1151, Q1157, Q1163, Q1170, Q1176, Q1185, Q1194]*
- Weight = W = mg; location ke hisaab se badalta hai kyunki g badalta hai. *[Q1151, Q1157, Q1163, Q1170, Q1176, Q1185, Q1194]*
- Moon par g = g_earth/6; isliye Moon weight = Earth weight / 6. *[Q1151, Q1157, Q1163, Q1170, Q1176, Q1185, Q1194]*
- 60 kg mass wale insaan ka mass Moon par bhi 60 kg hi rahega; weight ≈ 100N ho jaayega (Earth par ≈ 600N). *[Q1151, Q1176]*
- 5 kg mass wale object ka weight Moon par 1/6 ho jaata hai. *[Q1157, Q1163]*
- Mass ka SI unit kg; Weight ka SI unit Newton (N). *[Q1151, Q1157, Q1185, Q1194]*
- Mass scalar hai; Weight vector hai (direction = center of body ki taraf). *[Q1194]*
- Beam balance se mass measure hoti hai; Spring balance se weight measure hota hai. *[Q1170, Q1185]*
- Beam balance Moon par bhi same reading degi (mass same); Spring balance 1/6 reading degi (weight kam). *[Q1170, Q1185]*

2. Universal Law of Gravitation: Newton's Law

- Newton's Universal Law of Gravitation (1687): Har do bodies ek doosre ko attract karti hain force se jo unke masses ka product ke directly proportional aur unke beech ki doori ke square ke inversely proportional hota hai.

[Q1152, Q1158, Q1164, Q1171, Q1177, Q1186, Q1195]

- Formula: $F = G(m_1 m_2)/r^2$. *[Q1152, Q1158, Q1164, Q1171, Q1177, Q1186, Q1195]*
- Is law se explain hota hai: apple ka Earth ki taraf girna, Moon ka Earth ke around ghoomna, planets ka Sun ke around ghoomna. *[Q1158, Q1164, Q1171]*
- Gravitational force hamesha attractive hoti hai; repulsive kabhi nahi. *[Q1186]*
- Force do bodies ke beech line joining par act karti hai. *[Q1195]*
- Jab doori double hoti hai toh force 1/4 ho jaata hai (inverse square law). *[Q1177]*
- Jab dono masses double hoti hain toh force 4 times ho jaata hai. *[Q1164]*
- Scientist = Isaac Newton; book = Principia Mathematica (1687). *[Q1152, Q1195]*

3. Gravitational Constant G

- G = Universal Gravitational Constant; value = 6.67×10^{-11} Nm^2/kg^2. *[Q1159, Q1165, Q1172, Q1178, Q1187, Q1196]*
- G ka value universal hai — kahin bhi same (Earth, Moon, Mars, space sab jagah). *[Q1159, Q1165, Q1172, Q1178, Q1187, Q1196]*
- G ka value medium par depend nahi karti (air, water, vacuum sab mein same). *[Q1159, Q1178, Q1187]*
- G ka dimensional formula = $[M^{-1}L^3T^{-2}]$. *[Q1172, Q1196]*
- Henry Cavendish ne pehli baar G ka value experiment se measure kiya tha (1798). *[Q1165]*
- G bahut chhota hai isliye rozmarra ki chhoti objects ke beech gravitational force negligible hoti hai. *[Q1196]*

4. Free Fall & Acceleration due to Gravity (g)

- Free fall = jab koi body sirf gravity ke effect mein gir rahi ho (air resistance negligible ho). *[Q1153, Q1160, Q1166, Q1173, Q1179, Q1188, Q1197]*
- Acceleration due to gravity (g) = Earth ke gravitational field mein kisi body ka acceleration; value ≈ 9.8 m/s^2 (ya 10 m/s^2 approximate). *[Q1153, Q1160, Q1166, Q1173, Q1179, Q1188, Q1197]*
- g ka value Earth ke surface par sab jagah same nahi — equator par kam, poles par zyada. *[Q1166, Q1173, Q1179, Q1188]*
- g ka value altitude (height) par kam hota hai kyunki center se doori badhti hai. *[Q1153, Q1160, Q1166, Q1173, Q1188, Q1197]*
- g ka value depth mein bhi kam hota hai (Earth ke center tak jaane par zero ho jaata hai). *[Q1173, Q1188]*
- Formula: $g = GM/R^2$ (surface par). *[Q1179, Q1188, Q1197]*
- Formula height par: $g' = g/(1 + h/R)^2 \approx g(1 - 2h/R)$ jab $h << R$. *[Q1153, Q1160, Q1197]*
- Formula depth par: $g' = g(1 - d/R)$. *[Q1173, Q1188]*
- Sabhi bodies same height se same g se girte hain (Galileo's experiment) — mass, size, shape par depend nahi (air resistance nahi ho toh). *[Q1166, Q1179, Q1193, Q1197]*
- Feather aur coin vacuum mein saath girte hain; air mein alag kyunki air resistance alag. *[Q1179]*
- Free fall mein body ka weight zero feel hota hai (weightlessness). *[Q1197]*
- g depend karta hai: G (constant), M (Earth ka mass), R (Earth ka radius), height, depth — falling object ke apne mass par NAHI depend karta. *[Q1193]*

5. g ka Variation: Height, Depth, Latitude

- Equator par g kam ≈ 9.78 m/s^2; Poles par g zyada ≈ 9.83 m/s^2. *[Q1166, Q1173, Q1179, Q1188]*

- Reason: Earth oblate spheroid hai (equator par thodi chaudi) → equator par R zyada → g kam. *[Q1166]*
- Plus equator par centrifugal force maximum → g aur kam. *[Q1188]*

- g height (h) par: $g_h = g(R/(R+h))^2$ → height badhne par g kam. *[Q1153, Q1160, Q1197]*
- g depth (d) par: $g_d = g(1 - d/R)$ → depth badhne par g kam; center par g = 0. *[Q1173, Q1188]*
- Agar h = R (Earth ke radius jitni height) → $g_h = g/4$. *[Q1153, Q1160]*
- Agar d = R/2 (adhi depth) → $g_d = g/2$. *[Q1173]*
- g value maximum poles par; minimum equator par. *[Q1166, Q1188]*
- Body ko equator se North ya South pole tak le jaane par weight badhta hai (g poles par zyada hone ki wajah se), chahe kis pole ki taraf jaayein result same hota hai. *[Q1184]*

6. Weightlessness

- Weightlessness = jab koi object apna weight "feel" nahi karta; normal reaction zero ho. *[Q1154, Q1161, Q1167, Q1174, Q1180, Q1189, Q1198]*
- Astronauts space station mein weightless dikhte hain kyunki wo bhi Earth ke saath free fall mein hain. *[Q1154, Q1161, Q1167, Q1174, Q1180, Q1189]*
- Weightlessness ka matlab gravity nahi hai — aisa nahi; gravity toh hai lekin body aur container dono saath gir rahe hain isliye normal force nahi lagti. *[Q1161, Q1167, Q1174, Q1180, Q1189, Q1198]*
- Weightlessness mein:
  - Paani ke drops gol shape mein float karte hain.
  - Candle ka flame gol ho jaati hai.
  - Body ka blood circulation upar ki taraf shift hota hai (face swollen dikhta hai).
  - Astronauts ko exercises karni padti hain bone/muscle loss prevent karne ke liye.
- Weightlessness sirf space mein nahi — freely falling lift mein bhi feel hota hai. *[Q1154, Q1189]*
- Aircraft "Vomit Comet" parabolic flight mein weightlessness create karta hai training ke liye. *[Q1180]*

7. Escape Velocity

- Escape velocity (v_e) = minimum velocity jis se koi body Earth ki gravity se permanently bahar nikal jaaye. *[Q1155, Q1162, Q1168, Q1175, Q1181, Q1190, Q1199]*
- Formula: $v_e = \sqrt{2GM/R} = \sqrt{2gR}$. *[Q1155, Q1162, Q1168, Q1175, Q1181, Q1190, Q1199]*
- Earth ke liye $v_e \approx 11.2$ km/s (ya ≈ 11.2 km/sec). *[Q1155, Q1162, Q1168, Q1175, Q1181, Q1190, Q1199]*
- Escape velocity mass of projectile par depend nahi karti. *[Q1162, Q1175, Q1181, Q1199]*
- Escape velocity direction par depend nahi karti (kisi bhi direction mein same speed chahiye). *[Q1155, Q1168]*
- Escape velocity height se kam hoti hai jab object pehle se upar ho (kyunki gravity kam). *[Q1190]*
- Moon ka escape velocity ≈ 2.38 km/s (Earth se kam kyunki Moon ki mass aur radius kam). *[Q1168, Q1199]*
- Agar koi body 11.2 km/s se kam velocity se throw kare toh Earth ke around orbit mein ghoomegi ya wapas gir jaayegi. *[Q1155, Q1175]*
- Escape velocity = $\sqrt{2}$ × orbital velocity. *[Q1190]*

8. Kepler's Laws of Planetary Motion

- Kepler's First Law (Law of Orbits): Planets elliptical orbits mein ghoomte hain, Sun ek focus par hota hai. *[Q1156, Q1169, Q1175, Q1182, Q1191, Q1200]*

- Kepler's Second Law (Law of Areas): Planet-Sun line equal areas in equal times mein sweep karti hai → planets Sun ke paas tez, door slow. *[Q1156, Q1169, Q1175, Q1182, Q1191, Q1200]*
- Kepler's Third Law (Law of Periods): $T^2 \propto r^3$ (period ka square ∝ semimajor axis ka cube). *[Q1156, Q1169, Q1175, Q1182, Q1191, Q1200]*
  - Yaani $(T_1/T_2)^2 = (r_1/r_2)^3$. *[Q1169, Q1182, Q1191]*
- Example: Agar kisi planet ka distance 4 times ho toh period = 8 times (kyunki $T^2 = 4^3 = 64 \rightarrow T = 8$). *[Q1169]*
- Kepler ne ye laws Tycho Brahe ke observations se derive kiye. *[Q1156, Q1191]*
- Newton ne in laws ko apne gravitation law se mathematically prove kiya. *[Q1191]*
- Ye sirf planets ke liye nahi — koi bhi satellite ya moon ke liye valid hain. *[Q1200]*

9. Centripetal Force & Satellite Motion

- Satellite Earth ke around tabhi ghoom sakta hai jab gravitational force = centripetal force ho. *[Q1156, Q1169, Q1175, Q1182, Q1191, Q1200]*
- Centripetal force = $mv^2/r = m\omega^2 r$. *[Q1156, Q1182]*
- Gravitational force hi centripetal force provide karti hai satellite ke liye. *[Q1156, Q1175, Q1182, Q1200]*
- Satellite ki orbital speed: $v = \sqrt{GM/r}$. *[Q1156, Q1182, Q1191]*
- Satellite ka period: $T = 2\pi\sqrt{r^3/GM}$. *[Q1169, Q1182, Q1191, Q1200]*
- Higher orbit → lesser speed → greater period. *[Q1191, Q1200]*
- Geo-stationary satellite ka period = 24 hours (Earth ke rotation ke barabar) → height ≈ 36,000 km. *[Q1156, Q1182, Q1191, Q1200]*
- Polar satellites pole to pole orbit mein hain; Earth rotate karti hai isliye poori Earth cover ho jaati hai. *[Q1182]*
- Satellites ke uses: communication, weather forecasting, GPS, remote sensing. *[Q1182, Q1191]*

10. Potential & Kinetic Energy in Gravitation

- Gravitational potential energy (PE) = $-GMm/r$ (negative kyunki infinity par zero maana jaata hai). *[Q1162, Q1175, Q1181, Q1190, Q1199]*
- Kinetic energy (KE) = $\frac{1}{2}mv^2 = GMm/(2r)$ (circular orbit mein). *[Q1162, Q1190]*
- Total energy (TE) = PE + KE = $-GMm/(2r)$. *[Q1162, Q1190, Q1199]*
- Total energy hamesha negative hoti hai bound orbit ke liye. *[Q1162, Q1190, Q1199]*
- Binding energy = $|TE| = GMm/(2r)$. *[Q1190]*
- Escape karne ke liye body ko itni energy deni padti hai ki $TE \geq 0$ ho jaaye. *[Q1190, Q1199]*
- Gravity dwara kiya gaya work sirf vertical height difference par depend karta hai ($W = mgh$), horizontal position ya kinetic energy change par direct nahi (gravity vertically downward act karti hai). *[Q1183]*

11. Geostationary & Polar Satellites

- Geostationary satellite: *[Q1156, Q1182, Q1191, Q1200]*
  - Equatorial plane mein orbit.
  - Period = 24 hours (Earth rotation ke saath synchronous).
  - Height ≈ 35,786 km (≈ 36,000 km).
  - Fixed position dikhta hai ground se → communication ke liye best.

- Polar satellite: *[Q1182]*
  - Near-polar orbit mein.
  - Lower height (few hundred km).
  - Period ≈ 100 minutes.
  - Earth ke rotation ki wajah se poori surface cover kar leta hai.
  - Weather monitoring, remote sensing ke liye use hota hai.

12. Projectile Motion

- Horizontal throw karne par body projectile ki tarah move karti hai. *[Q1154, Q1189]*
- Horizontal velocity constant rehti hai (agar air resistance nahi); vertical velocity g se badhti hai. *[Q1189]*
- Range maximum tab hoti hai jab angle = 45° (same height se throw karne par). *[Q1198]*
- Maximum height = $(u^2\sin^2\theta)/(2g)$. *[Q1198]*
- Range = $(u^2\sin 2\theta)/g$. *[Q1198]*

13. Gravitational Potential

- Gravitational potential (V) = work done per unit mass to bring from infinity to that point. *[Q1172, Q1187, Q1196]*
- Formula: $V = -GM/r$. *[Q1172, Q1187, Q1196]*
- Unit = J/kg. *[Q1172, Q1196]*
- Negative sign ka matlab: gravity attractive hai, isliye infinity se laane mein work by gravity hota hai (negative work by external agent). *[Q1196]*
- Gravitational field intensity (I) = $F/m = GM/r^2 = g$. *[Q1187, Q1196]*
- Relation: $I = -dV/dr$. *[Q1187]*

14. Tides

- Tides = ocean water ka rise and fall; mainly Moon ki gravity se hoti hai. *[Q1174, Q1180, Q1189, Q1198]*
- Sun bhi tides cause karta hai lekin Moon zyada effect kyunki wo paas hai. *[Q1180, Q1198]*
- Spring tides = Sun, Moon, Earth ek line mein (new moon/full moon) → high tides zyada high, low tides zyada low. *[Q1180, Q1198]*
- Neap tides = Sun-Moon 90° angle pe (first/third quarter) → tides kam extreme. *[Q1180, Q1198]*
- High tide = water upar; low tide = water neeche. *[Q1174]*
- Ek din mein generally 2 high tides aur 2 low tides aate hain. *[Q1174, Q1198]*

15. Miscellaneous Concepts

- Newton's law of gravitation se Coulomb's law similar hai structure mein (both inverse square) lekin Coulomb mein charges hain aur repulsion bhi ho sakta hai. *[Q1195]*
- Gravitational force hamesha attractive; weakest fundamental force. *[Q1186, Q1195]*
- Gravity ka range infinite hai; har body har doosre body ko attract karti hai. *[Q1186, Q1195]*
- Earth apne center se sabhi objects ko same acceleration (g) deti hai; isliye mass different hone par bhi sab saath

girte hain. *[Q1179, Q1197]*

- Two objects different masses ke free fall mein ek saath girte hain kyunki F=ma → $a=F/m=GM/R^2=g$; m cancel ho jaata hai. *[Q1179, Q1197]*
- g ka value Jupiter par zyada hota hai (mass zyada) → weight zyada; Mars par kam → weight kam. *[Q1151, Q1163, Q1176]*
- Gravity waves = Einstein ne predict ki thi; 2015 mein LIGO ne detect kiya. *[Q1195]*
- Black holes mein gravity infinite hoti hai; escape velocity > c (speed of light). *[Q1199]*
- Agar gravitational force ki absence ho (jaise deep space mein) toh rocket ka thrust stable rehta hai kyunki gravity ko counteract karne ki zaroorat nahi hoti — rate of gas ejection constant rahe toh thrust bhi stable rehta hai. *[Q1192]*

16. ⚠ Traps / Formula Cheat Sheet (MCQ Q1151–Q1200)

- ⚠ Trap: Mass kahin bhi same rehti hai — Moon, Earth, Mars, space sab jagah; weight badalta hai. *[Q1151, Q1157, Q1163, Q1176, Q1185, Q1194]*
- ⚠ Trap: Moon weight = Earth weight / 6; mass same. *[Q1151, Q1157, Q1163, Q1170, Q1176, Q1185]*
- ⚠ Trap: Beam balance se mass measure hoti hai; Spring balance se weight. *[Q1170, Q1185]*
- ⚠ Trap: Newton ka law = $F = Gm_1m_2/r^2$; $G = 6.67\times10^{-11}$. *[Q1152, Q1158, Q1164, Q1177, Q1186, Q1195]*
- ⚠ Trap: G ka value universal aur constant hai — location/medium par depend nahi. *[Q1159, Q1165, Q1172, Q1178, Q1187, Q1196]*
- ⚠ Trap: g ka value sab jagah same nahi — equator < poles, height par kam, depth par kam; equator se pole jaane par weight hamesha badhta hai. *[Q1166, Q1173, Q1179, Q1184, Q1188]*
- ⚠ Trap: Free fall mein sab objects same acceleration g se girte hain — apne khud ke mass par depend nahi karta (air resistance nahi ho toh). *[Q1166, Q1179, Q1193, Q1197]*
- ⚠ Trap: Weightlessness = gravity nahi hai aisa nahi; body aur container dono free fall mein hain isliye normal reaction zero. *[Q1161, Q1167, Q1174, Q1180, Q1189, Q1198]*
- ⚠ Trap: Escape velocity Earth ke liye = 11.2 km/s; Moon ke liye ≈ 2.38 km/s. *[Q1155, Q1162, Q1168, Q1175, Q1181, Q1190, Q1199]*
- ⚠ Trap: Escape velocity mass of body par depend nahi karti. *[Q1162, Q1175, Q1181, Q1199]*
- ⚠ Trap: Kepler's Third Law = $T^2 \propto r^3$; $T \propto r^{(3/2)}$. *[Q1156, Q1169, Q1182, Q1191, Q1200]*
- ⚠ Trap: Geo-stationary satellite ka period = 24 hours; height ≈ 36,000 km. *[Q1156, Q1182, Q1191, Q1200]*
- ⚠ Trap: Higher orbit → lesser speed → greater period. *[Q1191, Q1200]*
- ⚠ Trap: Gravitational PE hamesha negative (bound system ke liye). *[Q1162, Q1190, Q1199]*
- ⚠ Trap: g height par: $g_h = g(R/(R+h))^2$; depth par: $g_d = g(1-d/R)$; center par g=0. *[Q1153, Q1160, Q1173, Q1188]*
- ⚠ Trap: Tides mainly Moon se hoti hain, Sirf Sun se nahi. *[Q1174, Q1180, Q1189, Q1198]*
- ⚠ Trap: Spring tides = full/new moon; Neap tides = first/third quarter. *[Q1180, Q1198]*
- ⚠ Trap: Gravitational potential unit = J/kg; field intensity unit = N/kg = m/s^2. *[Q1172, Q1187, Q1196]*
- ⚠ Trap: Gravity ka work sirf vertical height difference par depend karta hai, horizontal displacement par nahi. *[Q1183]*
- ⚠ Trap: Gravitational force ki absence mein rocket ka thrust stable rehta hai (gravity counteract karne ki zaroorat nahi). *[Q1192]*

17. Numerical: Acceleration due to Gravity — Planetary Variations

- Kisi bhi planet ki surface gravity ka formula $g = GM/R^2$ hota hai, jahan M planet ka mass aur R uski radius hai. *[Q1201, Q1206, Q1207, Q1208, Q1209]*

- Agar kisi planet ka mass $M/3$ ho aur radius $R/3$ ho, toh uske surface par $g' = 3g$ hoga; radius square mein hone ki wajah se $1/9$ denominator upar jaake 3 ban jaata hai. *[Q1201]*
- Hypothetical planet jisme mass $M/2$ aur radius $R/3$ ho, uska surface gravity $(9/2)g$ hoga; calculation mein $(1/3)^2 = 1/9$ numerator ke $1/2$ se multiply hokar $9/2$ ban jaata hai. *[Q1206]*
- Agar mass $M/2$ aur radius $R/2$ dono half kar diye jaayein, toh $g' = 2g$ aata hai; radius ke square hone ki wajah se 4 upar jaata hai jo 2 se divide hokar $2g$ deta hai. *[Q1207]*
- Agar mass $2M$ aur radius $2R$ dono double ho jaayein, toh $g' = (1/2)g$ hoga; mass double se $\times 2$ lekin radius double se $\div 4$, net effect $\frac{1}{2}$. *[Q1208]*
- Kisi object ka weight $W = GmM/R^2$ hota hai; mass 15 times aur radius 4 times wale planet par weight $(15/16)W$ ho jaata hai kyunki 15 numerator mein aur 16 (4^2) denominator mein aata hai. *[Q1209]*

18. Numerical: Variation of g with Height from Surface

- Surface se height h par gravity ka formula $g' = g \times [R/(R+h)]^2$ hota hai, jahan R planet ki radius hai. *[Q1203]*
- Agar $h = R/10$ (yaani radius ka 10%), toh $g' = g \times (10/11)^2 \approx g \times 100/121 \approx 0.826g$; 9.8 m/s^2 ke liye yeh approximately 8.1 m/s^2 aata hai. *[Q1203]*

19. Numerical: Gravitational Force and Distance Relation

- Newton ke universal law ke mutabiq gravitational force $F \propto 1/r^2$ hota hai, jahan r do bodies ke beech ki distance hai. *[Q1202]*
- Agar force $F/9$ ho jaaye toh distance 3 times ho jaati hai kyunki $F \propto 1/r^2$, toh $F/9 \propto 1/(3r)^2$; square root relation mein distance badhti hai. *[Q1202]*

20. Numerical: Gravity Effects — Weight and Jump Height

- Kisi planet ya moon par jump ki maximum height $h \propto 1/g$ hoti hai same initial velocity maan kar; kyunki $v^2 = 2gh$, toh g kam hone par height utni hi zyada hoti hai. *[Q1205]*
- Moon par gravity $g/6$ hone ki wajah se, agar Earth par 1.5 m ki jump ho toh Moon par wohi energy se $1.5 \times 6 = 9$ m ki jump lag sakti hai. *[Q1205]*

21. Numerical: Orbital Period under Modified Force Laws

- Agar gravitational force distance ki n-th power ke inverse se vary kare ($F \propto 1/R^n$), toh circular orbit ka time period $T \propto R^{(n+1)/2}$ hota hai. *[Q1204]*
- Derivation: centripetal force $mv^2/R = GMm/R^n$ se $v = \sqrt{GM/R^{n-1}}$; phir $T = 2\pi R/v$ rakhte hue simplify karne par exponent $(n+1)/2$ aata hai. *[Q1204]*

22. Numerical: Formula Cheat Sheet

- Surface Gravity: $g = GM/R^2$
- Height par Gravity: $g' = g \times [R/(R+h)]^2 = GM/(R+h)^2$
- Weight: $W = mg = GmM/R^2$

- Gravitational Force: $F = Gm_1m_2/r^2$
- Jump Height (same initial velocity): $h \propto 1/g$
- Orbital Period (general force $\propto 1/r^n$): $T \propto R^{(n+1)/2}$
- Normal gravity ke liye (n=2): $T^2 \propto R^3$ (Kepler ka Teesra Niyam)

23. ⚠ Traps / Key Differences Summary (Numerical Q1201–Q1209)

- ⚠ Trap: Mass aur radius dono double karne par g double nahi, balki half (1/2) ho jaata hai; mass ×2 = ×2, radius ×2 = ÷4, net ×½. *[Q1208]*
- ⚠ Trap: Mass half aur radius half karne par g same nahi rehta, balki 2g ho jaata hai. *[Q1207]*
- ⚠ Trap: Mass half aur radius one-third karne par answer (9/2)g aata hai, (5/2)g nahi; denominator mein $(1/3)^2$ = 1/9 upar jaake 9 ban jaata hai. *[Q1206]*
- ⚠ Trap: Height par g calculate karte waqt direct subtraction nahi karna; h = R/10 par g' ≈ 8.1 m/s^2 aata hai, 4.5 ya 8.9 nahi. *[Q1203]*
- ⚠ Trap: Weight nikaalte waqt sirf mass ratio mat dekhna; radius ka square denominator mein aata hai (15/16 W, na ki 15/4 W). *[Q1209]*
- ⚠ Trap: Force F/9 hone par distance 3 times hoti hai, 9 times nahi; inverse square ke hisaab se square root lena padta hai. *[Q1202]*